Tracking Semantic Evolution of Data Science Roles with Unsupervised Latent Space Clustering

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Abstract

With the advent of AI, automation of traditional workflows, as well as roles being redefined as the value of human labor capital fluctuates with uncertainty, it is important for us to quantitatively investigate whether the traditional roles offered in the domain of data science in the job market is still relevant new college graduates and early career professionals. Job titles such as data scientist, data analyst, or machine learning engineers, while does paint a traditional impression of the typical experiences or tech stacks required, it is uncertain as to whether these traditional impressions are misinformation that misguide college graduates when pursuing their career path in 2026, as well as whether students nowadays need to formulate a brand new understanding in these job function categories within data science, and plan accordingly to embrace the massive unseen changes in the kinds of talents that are actually being demanded by industries in today's market. To this end, we propose a framework that collects large volumes of unique LinkedIn job postings and apply an unsupervised semantic clustering pipeline based on sentence embeddings, manifold reduction, density-based clustering, generative cluster labeling, and temporal latent space stability analysis. Our own preliminary results show that the early-career market is not organized around one unified data science profession definition, but rather around many narrower and more specialized role groupings, some of which likely would not have been clearly discovered and is still in an emerging state. We further find that certain roles are consistently subjected to semantic change over time, while others remain comparatively stable. In addition, by measuring the internal description consistency over a large number of summaries of the same cluster job-description language, it is revealed that some clusters are internally inconsistent to begin with, meaning that the kinds of roles grouped under the same label can vary substantially by the company and team it is placed in. Overall, our findings suggest that the entry-level data science labor market is both more differentiated and more dynamic than what people conventionally imagine them to be.

1 Introduction

Data science is a new emerging profession, and one of the fastest growing fields. With it evolving quickly from a combination of different disciplines such as statistics, computer science, and business analytics, its structure can lack well-defined occupational boundaries. The term "data scientist" is a very vague term and can mean multiple things like machine learning, data cleaning and processing datasets, business analytics, data engineering, etc. In addition, job postings themselves can be unreliable sig-

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nals of the actual roles content, where similar titles may have completely different responsibilities and tasks. This ambiguity is especially amplified at the early-career level. Roles like "data science intern" and "junior data analyst" can refer to very different roles and expectations. For students and early career individuals hoping to enter the field, it can be extremely difficult to decipher which skills to build, what to expect, and most importantly what career paths realistically exist when this vagueness exists.

Prior work has touched briefly on the structure of data science, mostly at a broad level and focusing mostly on mid-to-senior-level positions. Furthermore, the data science field is constantly and actively evolving, meaning that a snapshot that was taken of roles years ago may not accurately reflect what is going on in the current landscape. In addition, no study has asked whether early-career roles are becoming more stable over time, a question that would be more actionable for students, educators, and early-career professionals.

This paper aims to address that gap by asking: "Are early-career data science roles splitting into distinct, stable "micro-professions" that are not visible from job titles or job descriptions alone?" In this paper, micro-professions refer to specialized subcategories of data science work that share similar skills, responsibilities, industry contexts, or business functions. To deep dive into this, we analyze a collection of 10,603 unique early-career job postings collected from LinkedIn on an hourly basis since December 2025, focusing on roles in the data science domain including data scientists, data analysts, data engineers, and machine learning roles. Instead of focusing on job titles alone, we use an unsupervised approach, allowing clusters to emerge naturally from the semantic sentence embeddings from the job descriptions. This allows us to identify role clusters on the content of the description rather than the labels that appear on the surface. We intentionally kept our filtering broad in order to capture niche roles that may not fit into already established categories and to potentially discover new definitions of data science roles. Tracking shifts in these cluster distributions over time allows us to study a more detailed and temporally aware view of how early-career data science roles are evolving, overall analyzing stability. The findings contribute practical insight for students who are trying to navigate this field.

2 Related Works

With the expanding modern labor market and technological change, online job postings have emerged as an important tool for studying how labor demand evolves. Rahhal et al. conducted a survey paper on over 100 studies, finding that there was a huge spike in research using job postings with machine learning techniques to identify trends, skills, and shifts in labor demands across various industries [7]. Building on this, Tzimas et al. built a full pipeline used to convert raw online job postings as insight into the labor market, demonstrating that extraction of job titles, experience, and occupational categories is feasible through NLP and across different sectors [8]. Howison et al. also reinforced this by using generative AI to extract information from unstructured job postings across 68 major occupational groups, and finding that the AI-extracted information was not statistically different from corresponding survey data [3]. Overall, these studies show how job postings can combine into a valuable dataset for finding patterns, tracking occupational demand, and observing job characteristics. Moving from a broader labor market scale into the data science market itself, we find how popular the field has been to study through using job postings because of how rapidly its market is growing along with how poorly defined the field remains. De Mauro et al. challenged the idea of data science being one coherent role by utilizing web-scraped job posts and text mining to identify four distinct Big Data job families [1]. The paper argues that "Data Scientist" is a loose umbrella term that describes a complex set of roles, and not just a single profession. Michalcyzk et al. used 25,000 data science related job postings along with a combination of n-grams, TF-IDF, and topic modeling to cluster these job postings into 6 distinct roles: Business User, Data Analyst, Data Scientist, Data Engineer, Software Ar-

chitect, and Software Developer [6]. While these findings are useful to establish that distinct roles do in fact exist in the data science realm, it fails to address whether these roles remain stable, or whether they will shift across time. Gunklach et al. builds on this idea, arguing that data science has moved past the "unicorn" idea, where one person is expected to do everything [2]. Using a literature review and 16,348 job postings, they identified 9 different roles in data science, including Business Analyst, Data Analyst, Applied Data Scientist, Research Data Scientist, Data Engineer, ML Engineer, Software Developer, and Data Science Architect. This paper is relevant as it shows increasing role differentiation, but just like Michalczyk et al., is also more focused on mapping the structure of the field overall than early-career roles. Through these papers we were able to find how online job postings can reveal meaningful insights and differences across roles in the data science market. The question of whether these distinctions also appear in internships and early-career opportunities still remain. Jaeger et al. offers relevant context as they analyze postings from an internship-specific website, and find meaningful patterns and occupational structure such as positions more closely tied to a specific occupation are more likely to be paid and employers are more likely to respond positively to individuals who had prior internship experience [4]. However, these findings are not specific to data science internship postings, and it is unknown whether these patterns hold. Alongside these findings, no study or research has directly studied early-career data science positions or internships, and whether they are splitting into distinct and stable micro-professions across time. The papers mentioned before have touched base with the structure of the data science field at a professional level, without analyzing these distinctions at the entry level. This paper aims to address this gap in research by focusing on finding micro-professions within early-career data science prepositions, as well as analyzing any temporal shifts in the postings.

3 Methodology

Our core framework treats each job posting as a semantic representation of role content and then analyzes the structure of that representation in an unsupervised manner. We first encode job descriptions with the `all-MiniLM-L6-v2` sentence-transformer model, producing dense embeddings that capture similarities in responsibilities, skills, and domain emphasis across postings. These embeddings are then projected into a lower-dimensional space with UMAP and clustered with HDBSCAN to recover latent role groupings directly from the posting language. To interpret the resulting clusters, we combine keyword-based summaries with LLM-assisted labeling, and to study whether these clusters remain stable over time, we evaluate month-to-month distributional change using Maximum Mean Discrepancy (MMD) alongside bootstrap-based internal description consistency analysis.

3.1 Data Collection and Pre-processing

The primary dataset D consists of job postings retrieved via a serverless data ingestion pipeline that scrapes hourly updates of job postings from LinkedIn. The dataset has been collected on an hourly basis since December 2025 up until the time that this paper is written. We let D_{init} denote the initial set of job postings that are extracted from the job board. To ensure we are targeting the correct domain and to filter out postings from irrelevant domains, we apply a boolean filter based on target keywords in the job title such as "data scientist", "machine learning", "data analytics" etc.

Furthermore, we perform extensive text preprocessing to remove syntactic noise that does not semantically represent the nature of the job. In this case, for each document $d \in D$, we remove boilerplate texts, statements, and application instructions that are embedded almost identically within each job posting requirements. After this process, our final workable corpus of texts can be defined as $D_clean = \{d_1, d_2, ..., d_N\}$, where $N = 10,603$ number of job postings.

3.2 Semantic Projection

To transform the unstructured text into a mathematically traceable format, we map each document d_i into a continuous vector latent space. We choose the $all-MiniLM-L6-v2$ sentence transformer model for this task, given how the transformer can semantically embed sentences and long phrases into vectors that can coherently represent a lengthy job description's latent meaning.

If we let $f_\theta : V^* \to \mathbb{R}^p$ to represent the encoder function with learned parameters θ that encodes a sequence of vocabulary tokens V to a dense vector embedding. The final output will be an embedding matrix $X \in R^{N \times p}$ with each row vector $x_i = f_\theta(d_i)$ with a final dimension of $p = 384$. The encoding process is also accelerated with batch processing on an NVIDIA A100 GPU in our experiments.

3.3 Manifold Reduction and Clustering

To directly cluster vectors with a dimension of R^{384} will incur the curse of dimensionality with distance metrics being uninterpretable as the ratio of the nearest to farthest data points approaches to one. To mitigate this, we first apply the Uniform Manifold Approximation and Projection (UMAP) dimensionality reduction technique to project X onto a lower dimensional manifold $Y \in \mathbb{R}^m$, where $m = 10$. We chose this number as a result of an extensive hyperparemeter grid search that identified that reducing the original embedding to a 10 dimensional vector produces optimal clustering results. In addition, UMAP is also chosen over Principal Component Analysis (PCA) is because it preserves both local and global non-linear topological structures, which linear methodologies such as PCA cannot do. UMAP is also considered to preserve the low dimensional structure of embeddings with much better overall quality and efficiency than techniques like Stochastic Numerical Estimator (SNE) as well as its variant, T-SNE. Therefore UMAP is chosen as the best dimensionality reduction tool above all.

Following of this step, we apply the Hierarchical Density-Based Spatial Clustering of Application with Noise (HDBSCAN) technique on the reduced space Y. We select HDBSCAN over centroid-based clustering algorithms like K-Means because the true number of the subcategories of data science roles is unknown, and the density of these categories is highly variable. Given this, HDBSCAN, which performs clustering by defining clusters as highly dense regions separated by sparse regions, can inherently create good quality clusters by classifying isolated points as noise rather than forcing them into invalid clusters.

Expanding upon the aforementioned hyper parameter tuning, we perform grid search over 81 configurations of UMAP and HDBSCAN parameters. The optimal configuration is selected by maximizing the Silhouette Coefficient S. For a sample i, this is defined as:

$$S(i) = \frac{b(i) - a(i)}{max(a(i), b(i))}$$

where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the mean nearest-cluster distance.

3.4 Cluster Formulation and Generative Labeling

Raw cluster assignments lack inherent semantic meaning. To assign each cluster with a humanly interpretable label, we treat all documents within a given cluster C_k as a single concatenated document of all the text corpus combined. We then apply the class-based Term Frequency-Inverse Document Frequency (c-TF-IDF) algorithm. The c-TF-IDF score for term (or a single vocabulary) t in cluster c is computed as:

$$W_{t,c} = tf_{t,c} \times log(1 + \frac{A}{f_t})$$

where $tf_{t,c}$ is the frequency of term t in class c, A is the average number of words per class, and f_t is the frequency of term t across all classes. This function extracts the **top 50 most distinctive tokens per cluster**. These token vectors are then passed to a large language model (GPT-4o-mini) via a structured few-shot examples prompt to synthesize and assign each cluster with a formal categorical label.

3.5 Temporal Stability Analysis

The final component of our analytical framework aims to quantify how the characteristics of these clusters evolve across a temporal observation window. Or, at a higher level, how has the job-description language, descriptors, or wording language changed across time. We treat the sequence of monthly subsets within a cluster as a time series and evaluate the stability using two types of metrics:

3.5.1 Distributional Divergence

To measure how much the actual content of a job category changes from one month to the next, we calculate the Maximum Mean Discrepancy (MMD) value. Instead of assuming a pre-defined probability distribution of these documents (i.e. assuming each description inside a cluster follows a Gaussian normal distribution, which is highly unlikely), we compute the Maximum Mean Discrepancy for the same cluster between consecutive months of which documents of the same category were recorded.

To understand this algorithm, one can consider the embedded job descriptions from a given month to be represented as a distinct spatial distribution within a high-dimensional vector space, with the subsequent month represented as a second distribution. The algorithm quantifies the global geometric divergence between these two distributions to evaluate whether the fundamental skill requirements of the role have drifted. It accomplishes this by computing the aggregate pairwise similarities within the first temporal subset and the similarities within the second subset, and then penalizing this sum by the cross-distribution similarities between the two temporal periods. If the latent job requirements have fundamentally transformed, the cross-distribution similarity decreases significantly, resulting in a substantially increased discrepancy score.

Hence, if we let P and Q be the embedding distributions for a cluster at month t and $t+1$, with samples $X = \{x_1, ..., x_n\}$ from one month and $Y = \{y_1, ..., y_m\}$ from a subsequent or future month, the squared MMD is estimated as the following equation:

$$\begin{aligned} MMD^2(X,Y) = &\frac{1}{n^2}\sum_{i,j} k(x_i, x_j) \\ &+ \frac{1}{m^2}\sum_{i,j} k(y_i, y_j) \\ &- \frac{2}{nm}\sum_{i,j} k(x_i, y_j) \end{aligned} \tag{1}$$

We utilize a Radial Basis Function (RBF) kernel $k(x,y) = \exp(-\frac{||x-y||^2}{2\sigma^2})$ to measure the distance between two points and assigns a similarity score. The kernel bandwidth σ that is used to control the convergence rate of the similarity score to zero (as two points move apart) is dynamically established via using the median of all pairwise Euclidean distances in order to allow the metric to better adapt to the local geometry of the cluster data. A high MMD score signifies a substantial topological shift in the types of roles being aggregated within that cluster.

3.5.2 Internal Description Consistency

While MMDs capture vector-space drift of the clustered job descriptions, it does not provide sufficient semantic interpretability as to what kind of the job-description language, or which part of the job-description language changed. Hence, we introduce a bootstrap summarization technique to capture the linguistic coherence between the clusters' job-description language across time. For each cluster at time t, we draw $N = 10$ number of independent ranodm samples of job descriptions. We utilize the LLM to generate a textual summary of each sample. These summaries are then embedded using the original sentence transformer to produce summary vectors $U = u_1, ..., u_B$. We then compute the mean pairwise cosine similarity score among these summary vectors:

$$\text{Sim}(U) = \frac{1}{\binom{B}{2}} \sum_{i<j} \frac{u_i \cdot u_j}{\|u_i\|\|u_j\|}$$

From this similarity score, a high positive value indicates that random subsets of the cluster yield

identical job-description language characteristics or descriptions of the same job role in their respective category. Whereas with a lower score indicates that the two random subsets are not similar, and reflects a change in how the job-description language is shaped for a specific category of data science role.

By tracking this semantic similarity matrix over consecutive months, we will be able to capture highly interpretable evidence of how specific data science job roles dynamically mature, change, or pivot in the labor market due to evolving demands for new skills and types of data science talents.

4 Experiments and Results

We evaluate the proposed framework on a final corpus of 10,603 unique early-career job postings collected between December 2025 and May 2026. After projecting the semantic embeddings into a lower-dimensional space and applying density-based clustering with the best hyperparameter configuration identified through grid search, the final model identifies 45 non-noise clusters with a silhouette score of 0.7028. For an unsupervised text clustering task of this scale, this score suggests that the embedding space captures meaningful distinctions among different types of early-career roles. At the same time, 5,395 postings, or 50.9% of the corpus, are labeled as noise. Rather than interpreting this only as a limitation, we view it as evidence that the early-career data science labor market contains substantial overlap between job types, mixed responsibilities across roles, and postings that do not fit neatly into one clearly defined occupational category.

4.1 Micro-Data Science Profession Clusters

The clusters that emerge from the model show that early-career data science jobs are not organized around one unified profession. In this paper, we use the term micro-professions to refer to specialized subcategories of data science work that share similar skills, responsibilities, industry contexts, or business functions. As shown in Figure 1, the embedding space groups postings into a large number of micro-professions that differ in their technical requirements, industry contexts, and business functions. Some clusters align with expected categories such as *Data Engineering Intern*, *Machine Learning Intern*, *Data Science Intern*, and *Business Intelligence Analyst*. However, other clusters reveal narrower and more domain-specific role categories, including *Mortgage Data Analyst*, *Biotechnology Intern*, *Political Research Analyst*, *Entertainment Data Analyst*, *Supply Chain Data Analyst*, and *Industrial Robotics Intern*. This suggests that even when the initial keyword filtering is broad, semantic representations can still uncover niche roles that are more specialized than their listed job titles may imply.

This result supports the hypothesis that early-career data science roles are splitting into distinct occupational subgroups. The market is not only differentiating between broad categories such as "data scientist" and "data analyst," but also between versions of these roles that are shaped by specific industries, products, and organizational needs. Therefore, job titles alone may understate the diversity of early-career data science work and may fail to capture the specific skill demands associated with different parts of the labor market.

4.2 Temporal Stability of Clusters

Next, we evaluate whether the discovered clusters remain stable over time by computing month-to-month Maximum Mean Discrepancy (MMD) scores within each cluster. Among clusters with sufficient monthly data, we observe substantial variation in temporal stability. As shown in Figure 2, the most volatile cluster is *Political Research Analyst*, with a mean MMD of 0.3637, followed by *Entertainment Data Analyst* at 0.1500, *Biotechnology Intern* at 0.1298, *Insurance Analyst* at 0.1214, and *Business Analyst Intern* at 0.1040. These higher values indicate that the language used to describe some roles changes considerably from one month to the next. This may reflect shifts in required skills, job responsibilities, or business contexts within

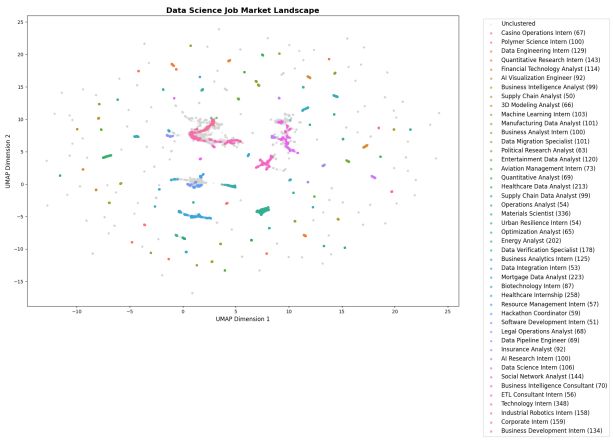


Figure 1: Vector embedding landscape of the identified early-career data science job clusters in two-dimensional embedding space. Each colored region corresponds to a discovered cluster, while the unclustered points represent postings whose semantic content does not map clearly to a stable or consistent role category.

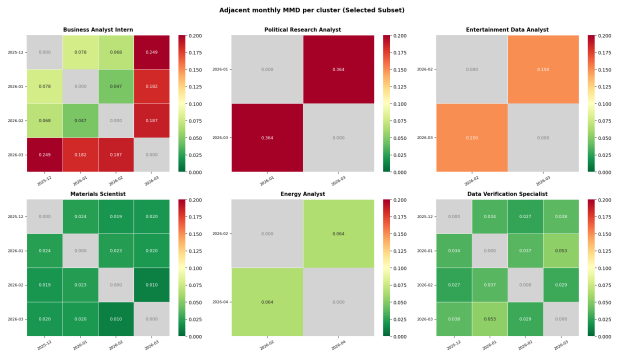


Figure 2: Distributional shift measured by Maximum Mean Discrepancy across monthly transitions for a focused subset of representative clusters. Warmer colors indicate larger changes in job-description language from one month to the next, while cooler colors indicate more stable descriptions of skills, responsibilities, and hiring expectations.

certain emerging data-science-related roles.

In contrast, the most stable clusters include *Materials Scientist*, with a mean MMD of 0.0190, *Corporate Intern* at 0.0321, *Data Verification Specialist* at 0.0335, *Healthcare Internship* at 0.0390, and *Business Development Intern* at 0.0406. These lower scores suggest that some role categories are described and recruited for in relatively consistent ways across time. Together, these findings show that the early-career data science job market is not evolving uniformly. Some niche data science functions appear to be stabilizing into repeatable patterns of skill requirements and job responsibilities, while others remain less clearly defined and continue to change as employers redefine their hiring needs.

4.3 Shifts in Job-Description Language Among High-Volatility Clusters

The most volatile clusters are not merely noisy or diverse in a random structure. Instead, they often reflect real changes in how employers define a role over time. For example, the *Political Research Analyst* cluster shifts from postings centered on political research support and data visualization in January 2026 to postings emphasizing graduate-level analytics research, methodology, and multilingual communication in March 2026. Similarly, the *Biotechnology Intern* cluster moves across several distinct descriptions of required skills and responsibilities. It begins with postings focused on commercial analytics and predictive modeling, then shifts toward protein generative modeling and technical machine learning research, and later moves toward clinical data science work involving the analysis of data in the domain of biological research, such as genes, proteins, or other measurable signals related to disease progression or treatment response. The *Business Analyst Intern* cluster also shows noticeable fluctuation in how the role is described, with monthly summaries ranging from intelligence-oriented data science roles, to database validation work, to database engineering and reporting, and later to descriptions that seeks applicants with seniority in experience in intelligence services.

As shown in Figure 3, these examples suggest that many early-career job categories are being reshaped by artificial intelligence, data automation, and broader labor-market changes. Even roles that have existed for a long time, such as



Figure 3: Subset visualization of the highest-volatility clusters across months in the embedding space. The shifting monthly positions illustrate how certain clusters change in their central job-description content over time rather than remaining fixed around one consistent definition of the role.

business analytics roles, may now involve changing expectations around technical skills, domain knowledge, and analytical responsibility. For early-career job seekers, this means that role categories with large month-to-month changes require closer attention. Students and applicants may need to track how these roles are changing over time rather than relying only on traditional job titles or older assumptions about what a role would typically involve.

4.4 Internal Description Consistency

In addition to measuring how cluster descriptions change over time, we also measure how internally consistent each cluster is across its individual job postings. To do this, we use a bootstrap-based procedure where we repeatedly draw random subsets of postings from the same cluster, summarize them using GPT-4o-mini, and measure how similar the resulting summaries are to one another. This step is important because while MMD scores can show whether a cluster changes across time, they do not reflect whether the postings inside the cluster are similar to one another in the first place. A cluster may be temporally stable across time but still contain a mix of unrelated or diverse job descriptions that are different against each other from the traditional sense. Therefore, the internal description consistency score helps evaluate whether postings within the same cluster describe similar skills, responsibilities, and hiring expectations. As shown in Figure 4, most clusters achieve relatively high internal description consistency scores. This suggests that the discovered clusters are not arbitrary groupings, but meaningful groups of job postings with similar descriptions of skills, responsibilities, and hiring contexts. The most internally consistent clusters are *Mortgage Data Analyst* (0.9647), *Business Intelligence Consultant* (0.9520), *Data Integration Intern* (0.9470), *AI Visualization Engineer* (0.9455), and *Data Pipeline Engineer* (0.9446). These high scores indicate that repeated random subsamples from the same cluster produce highly similar summaries. In other words, this suggests that employers describe these typical roles in relatively clear and consistent ways across the industry (i.e. roles in Business Analytics, Data Engineering, and Business Intelligence)

At the lower end, the least internally consistent clusters are *Resource Management Intern* (0.7866), *Business Analyst Intern* (0.8169), *Software Development Intern* (0.8432), *Biotechnology Intern* (0.8542), and *Data Verification Specialist* (0.8563). This is an interesting observation because several of these lower-consistency clusters also show high MMD values. In other

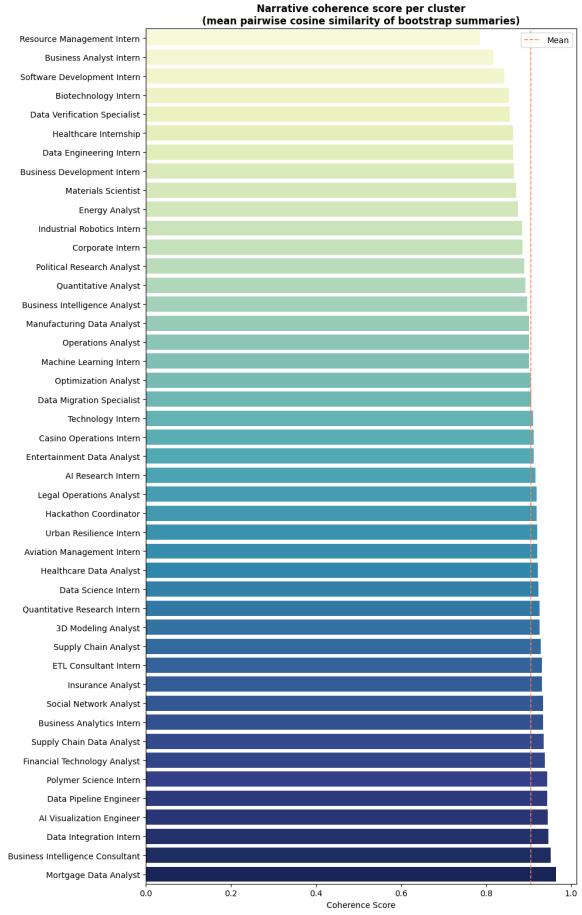


Figure 4: Internal description consistency scores for the discovered clusters. Higher values indicate that repeated random subsamples from the same cluster yield highly similar summaries, thereby suggesting they have stronger internal semantic consistency.

words, some of the categories that appear unstable across time are also less similar within the same cluster at a given point in time. The relationship between MMD and internal description consistency scores suggests that not all early-career data science roles are equally mature or equally well-defined. Some categories appear to have clear and recognizable job descriptions, while others may still be changing, internally mixed, or too general to represent one coherent job category.

Overall, these experimental findings show that early-career data science jobs are organized into many distinct micro-professions, or specialized subcategories of data science work with shared skills, responsibilities, and hiring expectations. However, these categories vary substantially in both the temporal stability and internal consistency of their job descriptions. While some roles already resemble clearly defined job categories, with consistent skill requirements and role descriptions being posted across time. Others remain volatile and continue to shift as employer demands evolve and adapt with the advent of new disruptive technology in the status quo. The main finding here is that the entry-level data science labor market contains more specialized categories and changes more over time than understandings of previous data science roles from the last decade would suggest.

5 Discussion

5.1 Early-Career Data Science is Already Fragmented

One immediate takeaway from our clustering results is that the early-career data science is more divided than the conventional job titles might suggest. We discovered 45 distinct semantic clusters, with expected roles like *Machine Learning Intern* and *Data Engineering Intern* all the way to domain-specific roles like *Mortgage Data Analyst*, *Political Research Analyst*, and *Biotechnology Intern*. This demonstrates how the early-career data science market has begun splitting into micro-professions that are not visible from looking at just the job titles alone. This finding supports our central hypothesis and expands on the work of Michalczyk et al. [6] and Gunklach et al. [2], who identified role differentiation at a mid-to-senior level, where now we found that it is meaningful at the entry level as well. Seeing that many of these roles are specialized is also surprising, seeing that these are entry level-positions and are often assumed to be interchangeable and undifferentiated. The results we have found suggest the opposite where even at the earliest stages of a data science career, employers are recruiting for roles that are within specific industries and technical stacks. This is an important discovery, as our once thought sin-

gle, unified data science career path may be misleading for students and early-career professionals.

5.2 Not All Roles Are Equally Mature

A key distinction in our findings is that role differentiation is not uniform throughout the market. Clusters such as *Mortgage Data Analyst* and *Business Intelligence Consultant* showcase high internal description consistency and low MMD scores, which suggests that these roles have already converged into stable, well-defined occupational identities. These roles therefore behave like well established professions, where their requirements, skill expectations, and job-description language framing are predictable as well as repeatable over time and across different employers. However, roles like *Political Research Analyst*, *Biotechnology Intern*, and *Business Analyst Intern* show high temporal volatility and lower internal description consistency, possibly indicating that employers themselves haven't reached a consensus on what these roles might require. We see that the distinction between stable and volatile clusters is important for early-career individuals because it suggests that some data science career paths are more predictable and plannable than others. For example, a student targeting to pursue a role such as *Data Pipeline Engineer* can expect consistent skill requirements and a more clear preparation roadmap, whereas a student targeting a less stable role like *Biotechnology Intern* might need to be ready to adapt and continuously update their skillset to match the requirements and expectations of the role. A question that can then be raised is whether the stability of the cluster is due to the age of the profession, the specificity of the industry it falls under, or the pace of technological change in that domain. Future work can investigate this problem further.

5.3 Macro Factors Are Driving Role Volatility

The job-description language shift analysis reveals that the volatility in high-MMD clusters is not simply due to random noise, but also reflects real and interpretable changes in what employers are looking for over time. The *Political Research Analyst* cluster shifted from data visualization support roles over to graduate-level multilingual analytics research. On the other hand, the *Biotechnology Intern* shifted from commercial analytics toward protein generative modeling and clinical data science. The *Business Analyst Intern* shifted in just a few months from intelligence-oriented data science roles, database validation work, database engineering and reporting, and eventually toward individuals with extensive experiences in the 'military intelligence' domain. These shifts show how forces like artificial intelligence and data automation are reshaping what some early-career roles mean in practice. Even with roles that have been in the industry for a long time, like Business Analytics, they are being redefined by the use of AI and automation tools. The pace these shifts are evolving are also worth looking into, where meaningful job-description language changes are detectable within a 5-month observation window, showing that the kinds of skill sets and expertise that early-career market demands for are changing extremely fast.

5.4 The Noise is Meaningful

With 50.9% of the postings being classified as noise by HDBSCAN, it is worth interpreting carefully. Instead of interpreting this amount of noise as a failure of the clustering approach, we argue that this high noise rate is a very informative finding within itself. It reflects the ambiguity and overlap that can make up the early-career data science market. Many job postings don't simply belong to any stable or repeatable role, where they might belong to a hybrid space between multiple role identities, or might even describe a role that is so novel that there is no cluster to fall into. However, the noise in our dataset may not be permanent with it being possible that as the field matures, these unclustered postings can fall into new stable micro-professions that are not visible yet in the current data. For students and educators, this is an important sig-

nal. A large amount of early-career data science opportunities cannot be mapped yet into a predictable career path, and navigating this unpredictable portion requires a different kind of preparation than the stable well-defined clusters require.

5.5 Coherence as a Proxy for Occupational Legitimacy

Another interesting finding from our internal description consistency analysis is how internal description consistency scores appear to track something like occupational legitimacy, which is the extent to which a role has been recognized and defined by the labor market. Clusters with high internal description consistency scores, like *Mortgage Data Analyst* and *Data Integration Intern* are roles that are stable and employers across a wide range of companies are basically telling the same thing about what the job involves. This suggests that there is a shared understanding of the role that has emerged in the market. On the other hand, clusters with low internal description consistency scores, such as *Resource Management Intern* and *Business Analyst Intern* are roles where different employers are telling different stories. This can mean that the role label is being used as a broad label with a poorly defined set of responsibilities. For job seekers, this might mean that low-coherence roles may lead to unpredictable interview experiences as well as mismatched expectations than what was originally expected, because the job content may differ greatly from employer to employer even when the job titles appear to be the same.

5.6 Limitations

There are several limitations that we want to acknowledge although our findings may be novel and informative. Firstly, we sourced our dataset solely from LinkedIn, which may not be representative of the full job market. This is especially true for roles that come from smaller companies or industries that only recruit through specialized job boards. As LinkedIn is a large platform, they can skew towards larger employers, underrepresenting niche or emerging professions that are outside of mainstream hiring platforms. Secondly, our observation window is from December 2025 to May 2026, which is a relatively short period of about five months. This may be a sufficient time to capture month-to-month shifts, but a larger span of multiple years would be needed to confirm whether the volatility we observed in this paper reflects seasonal hiring patterns or economic conditions, as well as longer structural changes in how roles are defined. Thirdly, the LLM cluster labeling that we used can provide interpretable and human-readable role names, but can also introduce a level of subjectivity in the clusters being characterized. Using different modeling versions or prompting strategies might produce different labels for the same cluster. This is why we make it known that the labels themselves should be understood as interpretive summaries rather than ground truth role categories. Fourth, although our focus is on early-career postings, it may have missed out on some relevant roles that do not contain the target keywords but still represent early-career data science work. Lastly, the noise rate of 50.9% which is relatively high means that our findings are based on approximately half of the postings, with the unclustered portion being underexplored in this paper.

5.7 Implications and Future Work

Despite the limitations, the findings that we have found have meaningful and practical implications. Looking from a students perspective, the finding that stable micro-professions suggests that rather than having a broader generalist preparation for the job market, a targeted approach may be a more effective career strategy. Furthermore, especially understanding which clusters are stable vs. volatile could potentially help students in making informed decisions such as which skills to invest in and the industries to target. Looking at future work, we can apply this framework to longer time horizons, a broader job board dataset, or other emerging fields like AI engineering or digital health to com-

pare whether the micro-professions that we find here can generalize more broadly. In addition, future studies can further dive into whether being in a stable and more well-defined role can lead to better outcomes for early-career individuals, such as higher pay or higher job satisfaction when compared to a vague and unstable role.

6 Conclusion

In this work, we investigated whether early-career data science roles are still accurately represented by traditional notions of roles in this domain, as well as by the conventional job titles that these roles are assigned with (i.e. Data Scientists, data analyst, or data engineer). And analyzed whether the job market has started splitting into more specific micro-professions that are not visible from these titles and conventional understandings alone. To this end, we proposed and tested an unsupervised framework that embeds job descriptions into the latent space and performs density-based clustering to discover unique job categories as well as measure the temporal distributions of those clustering to uncover insights about the coherence and changes in the job descriptions in each category. Our results suggest that while some early-career data science roles are already showcasing relatively mature and well-defined semantic job category embeddings, numerous other roles show much higher volatility, meaning that the skills, responsibilities, and expectations associated with them can change substantially within only a few months. This is a significant insight as both students and early career professionals often rely on broad job titles when deciding what skills to build or which roles to target in application. As our findings have shown, these titles may not provide adequate information about what the role actually requires, nor does the information that one receives from previous years (or even months) may align with the updated needs from the industry in regards to that specific role. Overall, this work should be understood as a proposal of framework towards understanding the semantic drifts of job descriptions over time for roles in the early career data science job market. Our results are solely based off one data source that is not comprehensive nor reflective of the entire US data science job market. Therefore the patterns that we identify should be only interpreted as a preliminary evidence of how this framework can reveal hidden structure and temporal change in early career job posting descriptions rather than as a final conclusion about the current labor market.

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A Appendix

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